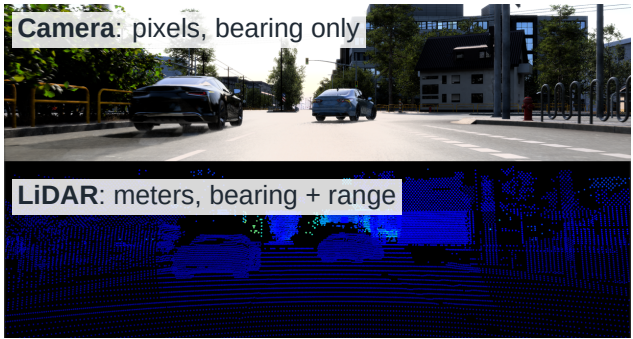
Coherent Fusion of LiDAR Geometry and LiDAR Intensity Images

Uncertainty propagation for intensity-image feature residuals

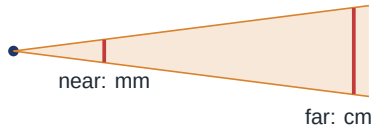
Wenda Zhao

UNIVERSITY OF TORONTO

LiDAR–Camera Fusion: Two Different Scales



1 pixel of error = ? meters

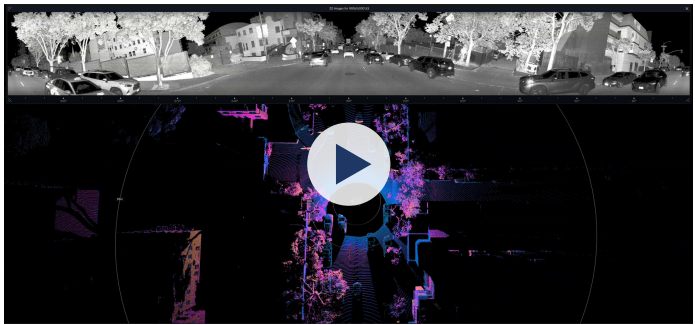


- Depends on **unknown depth**
- Plus extrinsic, time-sync errors

⇒ **Weights are tuned by hand**

UNIVERSITY OF TORONTO

Key Insight: The Intensity Image Is the LiDAR Scan



Top: intensity image. Bottom: point cloud. Same scan, same sensor.

Each pixel (u, v) also has a **range** ρ:

$$\mathbf{z} = \begin{bmatrix} u \\ v \\ \rho \end{bmatrix} \leftrightarrow \mathbf{p} = f^{-1}(\mathbf{z})$$

One feature = one LiDAR point

Same sensor, same noise model.

UNIVERSITY OF TORONTO

Back-Projecting a Feature to 3D

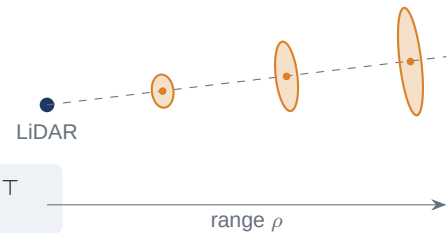
Image coordinates and range of feature i:

$$\mathbf{z}_i = \begin{bmatrix} u_i \\ v_i \\ \rho_i \end{bmatrix}, \qquad \mathbf{\Sigma}_{\mathbf{z}_i} = \begin{bmatrix} \mathbf{\Sigma}_{uv,i} & \mathbf{0} \\ \mathbf{0}^\top & {\sigma_{\rho,i}}^2 \end{bmatrix}$$

Back-project:

$$\mathbf{p}_i = f^{-1}(\mathbf{z}_i), \qquad \mathbf{\Sigma}_{\mathbf{p}_i} = \mathbf{J}_{f^{-1},i}\, \mathbf{\Sigma}_{\mathbf{z}_i}\, \mathbf{J}_{f^{-1},i}{}^\top$$

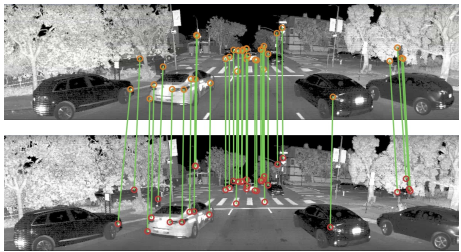
$\mathbf{J}_{f^{-1},i} = \partial f^{-1}/\partial \mathbf{z}_i$; $\mathbf{\Sigma}_{uv}$: learnt; σ_ρ^2: range noise



Across the beam: grows with range.
Along the beam: set by σ_ρ.

UNIVERSITY OF TORONTO

Residual 1: 2D Intensity-Image Feature Residual



Matched features: i at t_1, j at t_2

Lift i to 3D, move it to t_2, reproject:

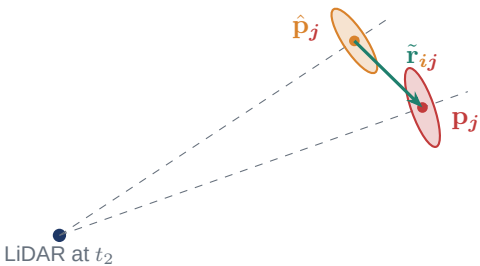
$$\hat{\mathbf{p}}_j = \mathbf{R}\,\mathbf{p}_i + \mathbf{t}, \qquad \mathbf{r}_{ij} = \mathbf{y}_j - f(\hat{\mathbf{p}}_j) \in \mathbb{R}^2$$

$$\mathbf{y}_j = [u_j, v_j]^\top, \qquad \mathbf{J}_{f,j} = \partial f / \partial \hat{\mathbf{p}}_j$$

$$\boldsymbol{\Sigma}_{\mathbf{r}_{ij}} = \underbrace{\boldsymbol{\Sigma}_{uv,j}}_{\text{observed } j} + \underbrace{\mathbf{J}_{f,j}\mathbf{R}\,\mathbf{J}_{f^{-1},i}\,\boldsymbol{\Sigma}_{\mathbf{z}_i}\,\mathbf{J}_{f^{-1},i}{}^\top\mathbf{R}^\top\mathbf{J}_{f,j}{}^\top}_{\text{predicted from } i} + \underbrace{\boldsymbol{\Sigma}_{ij}{}^{\text{corr}}}_{\text{matching}}$$

UNIVERSITY OF TORONTO

Residual 2: 3D Point-to-Point Feature Residual



The match j has a range too, so lift it:

$$\tilde{\mathbf{r}}_{ij} = f^{-1}(\mathbf{z}_j) - \hat{\mathbf{p}}_j \in \mathbb{R}^3$$

Each feature is a **3D point with a descriptor**: matches give point-to-point pairs without nearest-neighbour search.

$$\boldsymbol{\Sigma}_{\tilde{\mathbf{r}}_{ij}} = \underbrace{\mathbf{J}_{f^{-1},j}\,\boldsymbol{\Sigma}_{\mathbf{z}_j}\,\mathbf{J}_{f^{-1},j}^{\top}}_{\text{observed } j} + \underbrace{\mathbf{R}\,\mathbf{J}_{f^{-1},i}\,\boldsymbol{\Sigma}_{\mathbf{z}_i}\,\mathbf{J}_{f^{-1},i}^{\top}\mathbf{R}^{\top}}_{\text{predicted from } i} + \underbrace{\boldsymbol{\Sigma}_{ij}^{\text{corr}}}_{\text{matching}}$$

UNIVERSITY OF TORONTO

Two Residuals, One Noise Model

	2D feature residual	**3D point-to-point residual**
Residual	$\mathbf{y}_j - f(\hat{\mathbf{p}}_j)$ (pixels)	$f^{-1}(\mathbf{z}_j) - \hat{\mathbf{p}}_j$ (meters)
Range used	frame t_1	frames t_1 and t_2
Pose Jacobian	$-\mathbf{J}_{f,j}\,\partial\hat{\mathbf{p}}_j/\partial\boldsymbol{\xi}$	$-\partial\hat{\mathbf{p}}_j/\partial\boldsymbol{\xi}$ (as in ICP)
Observed side	$\boldsymbol{\Sigma}_{uv,j}$	$\mathbf{J}_{f^{-1},j}\boldsymbol{\Sigma}_{\mathbf{z}_j}\mathbf{J}_{f^{-1},j}{}^{\top}$

Both are built from the same $\boldsymbol{\Sigma}_{\mathbf{p}} = \mathbf{J}_{f^{-1}}\boldsymbol{\Sigma}_{\mathbf{z}}\mathbf{J}_{f^{-1}}^{\top}$ and the same LiDAR noise.

UNIVERSITY OF TORONTO

Coherent Fusion without Hand-Tuned Weights

LiDAR + camera

- Two sensors, two noise models
- Pixel-to-meter scale is unknown
- Relative weight tuned until it works

LiDAR + intensity image

- One sensor: one feature = one point
- Every residual covariance is propagated from $\mathbf{\Sigma}_{uv}$ and σ_ρ^2
- Weights follow from the noise model

$$\left(\mathbf{H}_g + \mathbf{P}_d \mathbf{H}_I \mathbf{P}_d\right) \delta\mathbf{x} = \mathbf{b}_g + \mathbf{P}_d \mathbf{b}_I, \qquad \mathbf{H}_I = \sum \mathbf{J}_\xi^\top \mathbf{\Sigma}_\mathbf{r}^{-1} \mathbf{J}_\xi$$

$\mathbf{P}_d$: projector onto the geometrically degenerate directions

UNIVERSITY OF TORONTO

Takeaways

- **Gap:** LiDAR–camera fusion has no common scale, so weights are hand-tuned.
- **Insight:** in a LiDAR intensity image, one feature is one LiDAR point.
- **Derivation:** propagate $\mathbf{\Sigma_z} = \mathrm{diag}(\mathbf{\Sigma}_{uv}, \sigma_\rho^2)$ into a 2D feature residual and a 3D point-to-point residual.
- **Result:** LiDAR geometry and intensity features fuse coherently, with no tuned weight.

UNIVERSITY OF TORONTO